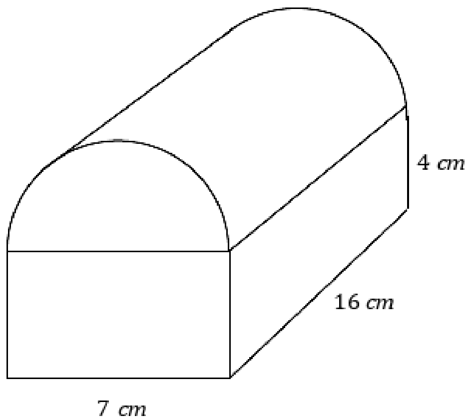
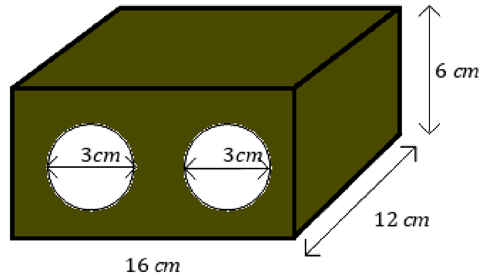
Volume of composite solids

1 Find the volume of the following solids, rounding your answers to one decimal place:

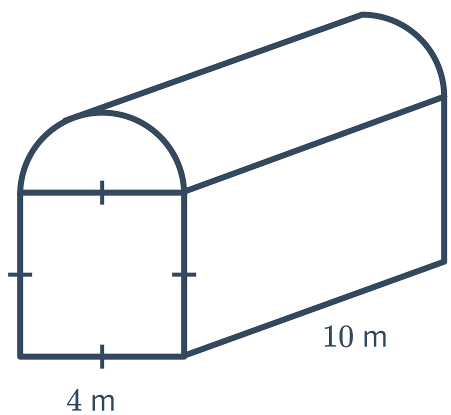
a



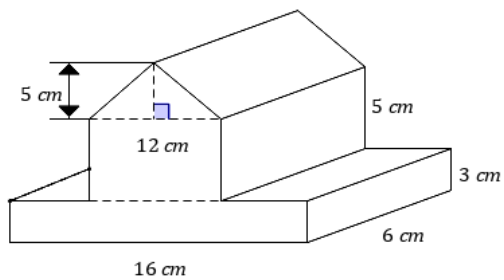
b



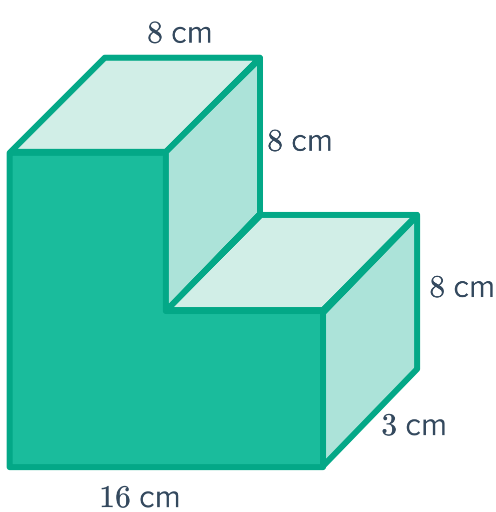
c



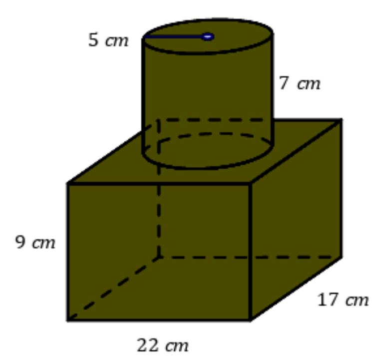
d



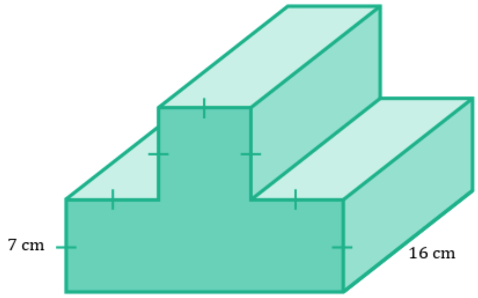
e

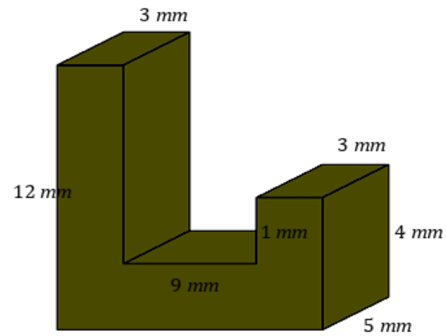


f

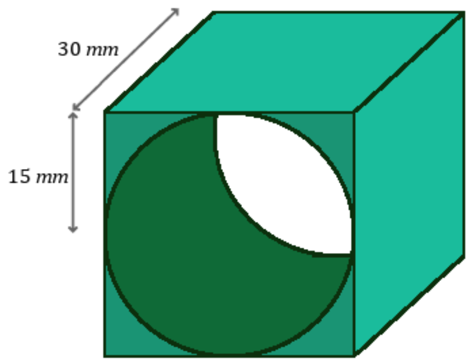


g

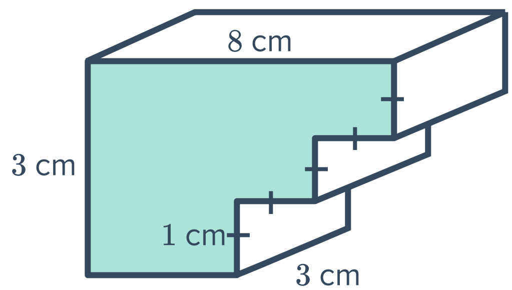




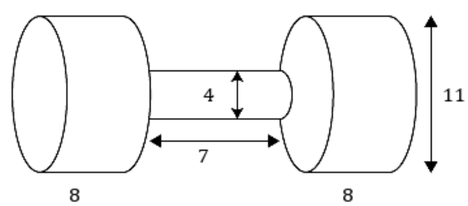
i



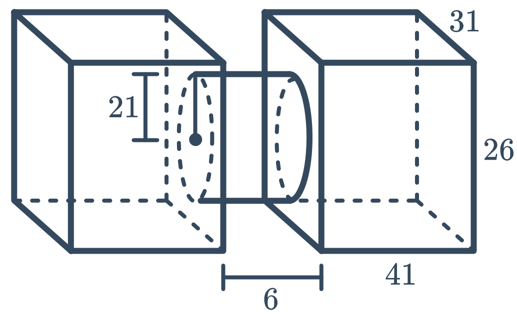
j



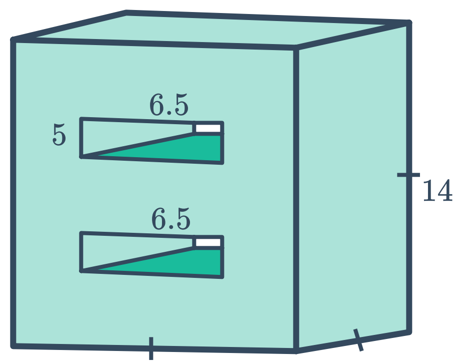
2 In the given diagram, the two cylinders at the ends are identical.
Calculate the volume of the solid, rounding your answer to one decimal place.
Note that all measurements are in centimeters.



3 In the given diagram, the two rectangular prisms are identical.
Find the volume of the composite solid, rounding your answer to two decimal places.



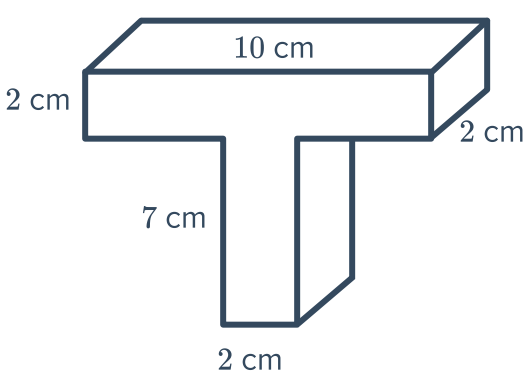
4 In the following diagram, the larger prism has had two identical holes carved out of it, each of which is a rectangular prism.
Find the volume of the remaining solid, rounding your answer to two decimal places.
Note that all measurements are in meters.



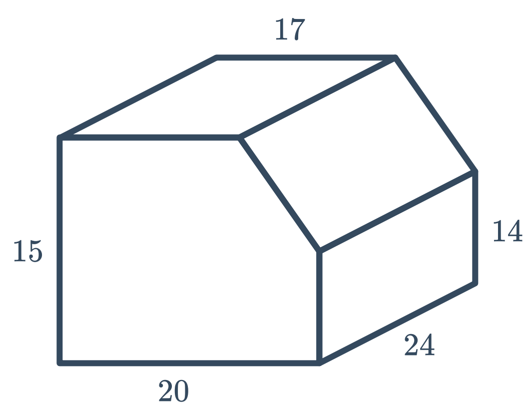
Additional questions

5 Find the volume of the following prisms:

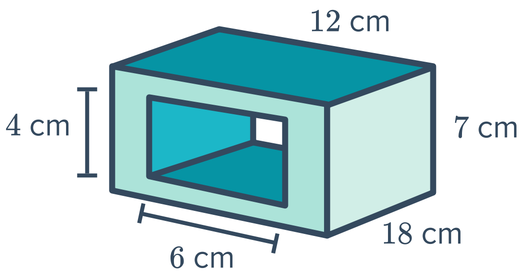
a



b

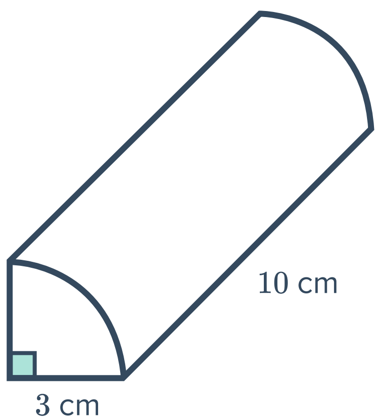


c

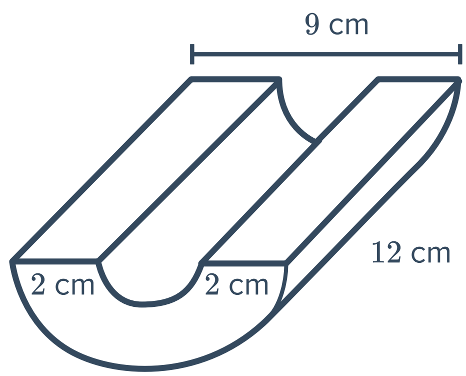


6 Find the volume of the following solids, rounding your answers to one decimal place:

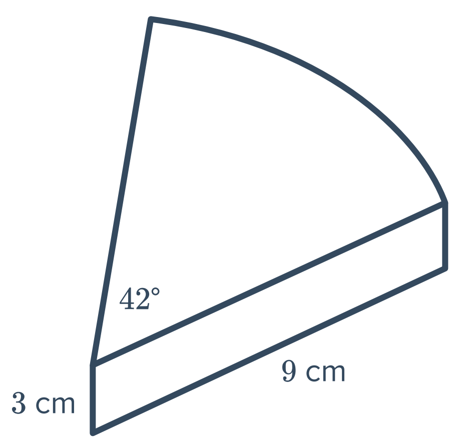
a



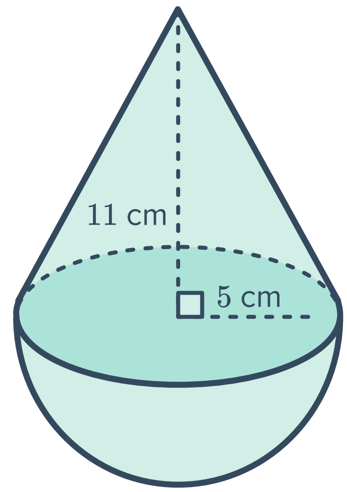
b



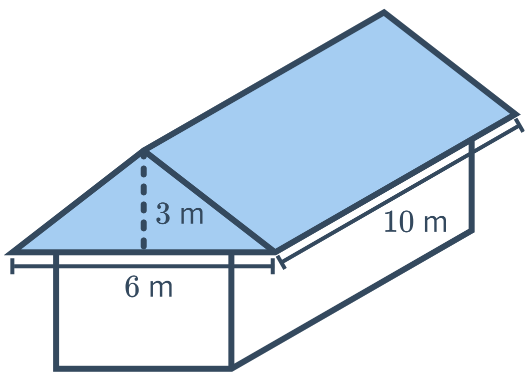
7 The following solid is $3\ \text{cm}$ thick.
Calculate the volume of the solid correct to one decimal place.



8 Find the volume of the composite figure shown.
Round your answer to two decimal places.



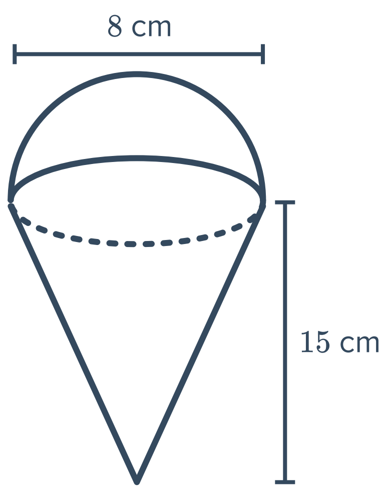
9 The roof of a cottage has the shape of a triangular prism with the dimensions shown below.
Find the volume of the roof.



10 The ice-cream cones at an ice-creamery have the dimensions indicated in the diagram.

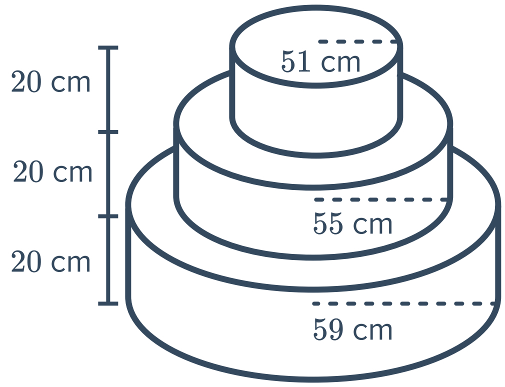
a How many milliliters of icecream can fit in each cone, including the hemisphere scoop on top? Round your answer to the nearest milliliter.

b The ice cream is bought in 10 L tubs. How many whole cones can be made with a single tub of ice-cream?



c Double cones are served with a second hemispherical scoop of the same dimensions as the first scoop.
How many whole double cones can be made with from a 10 L tub?

11 A wedding cake with three tiers is shown. The layers have radii of $51\ \mathrm{cm}$, $55\ \mathrm{cm}$ and $59\ \mathrm{cm}$.
If each layer is $20\ \mathrm{cm}$ high, calculate the total volume of the cake in cubic meters. Round your answer to two decimal places.



12 A cylindrical tank with diameter of $3\ \mathrm{m}$ is placed in a $2\ \mathrm{m}$ deep circular hole so that there is a gap of $40\ \mathrm{cm}$ between the side of the tank and the hole. The top of the tank is level with the ground.

a What volume of dirt was removed to make the hole? Give your answer to the nearest cubic meter.

b Find the capacity of the tank to the nearest liter.